Transductive Few-Shot Learning With High-Order Representation for Cross-Domain Hyperspectral Image Classification

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***Abstract*—Cross-domain hyperspectral image classification (HSIC) under few-shot conditions remains a formidable challenge due to severe data scarcity and pronounced cross-domain discrepancies. Recent advances in transductive few-shot learning (TFSL) offer a promising paradigm for tackling this problem. However, effective feature encoding, which is fundamental to TFSL, remains challenging for HSIs due to their complex high-order spectral–spatial correlations. Moreover, most existing TFSL methods assume a uniform class prior over query samples, which introduces severe bias under domain distribution mismatch and class imbalance, leading to significant performance degradation. To this end, we propose a novel method termed TFSL with high-order transformer-few shot learning (HOT-FSL), which mainly consists of three key components: high-order transformer encoder (HOTE), prototype-conditioned gated interaction module (PCGI), and conditional transport-based prototype-oriented unbiased transfer (CTPUT). The proposed HOTE employs factorized high-order attention with Kronecker-structured formulations to jointly model spatial–spectral dependencies and long-range interactions in HSI. PCGI models class prototypes from the support set and adaptively injects them into query features via contrastive routing, thereby enhancing feature discriminability and class consistency. During inference, CTPUT employs conditional transport (CT) with forward–backward navigators and an expectation–maximization (EM)-based solver to iteratively learn unbiased class statistics and refine prototypes under imbalanced query distributions. Comprehensive experimental results on three widely used benchmarks consistently validate the superior performance of the proposed method. The code of this article is available at https://github.com/1546342746/HOT-FSL**

***Index Terms*—Conditional transport (CT), high-order transformer, hyperspectral image classification (HSIC), metafeature interaction, transductive few-shot learning (TFSL).**

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I. Introduction

INTELLIGENT interpretation of remote sensing data has emerged as a central research topic, underpinning a wide range of critical downstream applications [1], [2]. Among the available sensing modalities, hyperspectral images (HSIs) are particularly distinctive for capturing tens to hundreds of contiguous narrow spectral bands per pixel, providing rich spectral signatures that enable precise material discrimination [3], [4]. Leveraging these properties, hyperspectral image classification (HSIC) assigns semantic labels to pixels [5], [6] and has been widely applied in areas such as precision agriculture [7], urban planning [8], and environmental monitoring [9].

Driven by the remarkable representational power of deep learning, the paradigm for HSIC has shifted from handcrafted feature design to end-to-end feature learning with deep neural networks. Convolutional neural networks (CNNs) [10] represent a fundamental class of deep learning architectures that exhibit significant performance advantages in HSIC [11]. Specifically, 1-D convolutions [12] are primarily designed to extract spectral dependencies, 2-D convolutions [13] to capture spatial context, and 3-D convolutions [14] to jointly model spectral–spatial correlations for comprehensive feature representation. Despite achieving excellent performance, these CNN-based methods are still difficult to model long-range dependencies due to the fixed receptive field of the CNN. Some studies [15], [16], [17] have alleviated this problem by continuously deepening network layers, but this remains insufficient for capturing long-range dependencies. Fortunately, the advent of visual transformers (ViTs) [18] provides a novel and effective approach for tackling these limitations [19], [20], [21]. While ViT-based methods effectively address the limitations of long-range dependence modeling, they do so at the cost of significantly increased computational complexity. Recently, state space models (SSMs) have emerged as promising architectures. Among them, Mamba [22] stands out for its ability to effectively balance computational efficiency and global contextual modeling for HSIC [23], [24], [25].

Although the aforementioned methods have achieved promising results, they are all supervised and, thus, heavily depend on large-scale labeled datasets. However, in real-world scenarios, HSIs are typically captured by satellite-

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or unmanned aerial vehicle (UAV)-mounted sensors, producing high-resolution data that are extremely expensive and time-consuming to annotate at scale. Therefore, achieving satisfactory performance under sample-scarce conditions better reflects the practical challenges faced in real-world HSIC applications. Unfortunately, empirical evidence has revealed that these supervised approaches, though demonstrating remarkable performance in conventional benchmarks, exhibit markedly degraded performance when applied to sample-scarce scenarios [26], [27]. Consequently, semisupervised [28] and self-supervised [29] learning paradigms have attracted considerable attention as effective means to alleviate the detrimental effects of label scarcity on classification performance. For instance, Sellami et al. [30] integrated adaptive dimensionality reduction with a semisupervised 3-D CNN to mitigate spectral redundancy and label scarcity in HSIC. Wu and Prasad [31] proposed a semisupervised deep learning framework based on CNN, where abundant unlabeled samples with pseudolabels are leveraged to enhance HSIC performance. Yue et al. [32] introduced a self-supervised learning method with adaptive distillation that leverages spatial–spectral similarity among unlabeled samples. Feng et al. [33] proposed a self-supervised divide-and-conquer generative adversarial network that leverages unlabeled samples to alleviate mode collapse. Moreover, metalearning [34] has emerged as a promising yet highly challenging paradigm that has attracted increasing attention in recent years. Often referred to as "learning to learn," it aims to leverage knowledge accumulated from multiple tasks to enable rapid adaptation to new tasks. By introducing an episodic learning strategy that mimics the few-shot setting during training, metalearning endows models with strong generalization capability. Few-shot learning (FSL) [35] represents a representative framework built upon the metalearning paradigm, and a growing body of work has pioneered the exploration of FSL for HSIC under label-scarce conditions. For example, Xi et al. [36] developed a class-covariance metric-based few-shot framework to enhance generalization and decision boundary flexibility in HSIC. Song et al. [37] proposed an FSL framework with label smoothing and metric space optimization to enhance the robustness of HSIC under limited labeled samples. Wang et al. [38] proposed a heterogeneous FSL framework that transfers knowledge from natural image datasets to improve classification performance with limited labeled samples.

Although these methods alleviate the issue of label scarcity to some extent, they typically rely on an implicit assumption that the training and testing data are drawn from the same or highly similar distributions. However, in practical remote sensing applications, HSIs are often acquired by different sensors, under diverse imaging conditions, or across distinct geographical regions, which inevitably leads to significant distribution shifts between the source and target domains. Under such circumstances, even when abundant unlabeled samples are available in the target domain, directly transferring semisupervised, self-supervised, or FSL models may still suffer from severe performance degradation. Consequently, the joint challenge of domain discrepancy and extremely limited labeled samples emerges as a more demanding yet practically relevant research problem. Fortunately, the rapid progress of cross-domain FSL (CDFSL) [39] has opened a promising avenue for addressing such scenarios since it permits the source and target data to come from heterogeneous domains and label spaces.

Owing to its effective learning paradigm, CDFSL has been widely adopted to address few-shot HSIC. First, domain adaptation constitutes a critical challenge for applying CDFSL to HSIC, and several studies have focused on effectively addressing this issue. For example, Ye et al. [40] introduced ADAFSL, an adaptive domain-adversarial FSL method that explicitly accounts for classwise conditional distribution discrepancies in HSIC. Cheng et al. [41] approached cross-domain few-shot HSIC from a causal perspective by proposing a causal metatransfer learning framework to disentangle causal and noncausal factors. Feng et al. [42] proposed a CDFSL framework based on decoupled knowledge distillation to address severe domain shift in HSI by jointly performing metaknowledge extraction and debiasing within a unified distillation paradigm. In addition, graph convolution [43] is also introduced because it can aggregate spectral spatial information across adjacent pixels, thereby enhancing feature robustness and class separability. For instance, Ye et al. [44] proposed a CDFSL framework based on graph convolution contrast (GCC-FSL) to transfer knowledge from source to target domains and mitigate domain shift in HSIC. Zhang et al. [45] proposed Gia-CFSL, a graph information aggregation CDFSL framework that leverages nonlocal spatial relationships to improve HSI under domain shift. However, these methods rely solely on supervised utilization of support samples, which partially limits the model's representational capacity under few-shot conditions. Recently, transductive FSL (TFSL) [46], [47] has gained increasing attention during the inference stage of FSL, as it leverages both support and query samples in a semisupervised manner to obtain more robust feature representations. Xi et al. [48] were the first to introduce TFSL into HSIC and proposed TEFSL, a CDFSL framework that integrates enhanced spectral–spatial embedding with graph-based prototype refinement.

Although existing methods have achieved certain successes, several critical challenges remain. First, effective feature extraction is particularly crucial, yet most prior approaches rely on general-purpose pretrained feature extractors. While these extractors perform reasonably on some datasets, they are not specifically designed for hyperspectral data and, therefore, struggle to capture the complex high-order spectral–spatial interactions inherent in HSIs. In addition, conventional prototype-based methods generate class prototypes solely from support sets and employ a fixed query–prototype matching strategy. Such a static approach is vulnerable to limited sample sizes, often producing noisy prototypes and preventing query features from being effectively adjusted, which leads to unreliable predictions. Moreover, HSI datasets frequently exhibit severe class imbalance, but most existing methods assume uniform class distributions in the query set. Under such conditions, this assumption introduces systematic bias, resulting in substantially degraded performance for minority classes.

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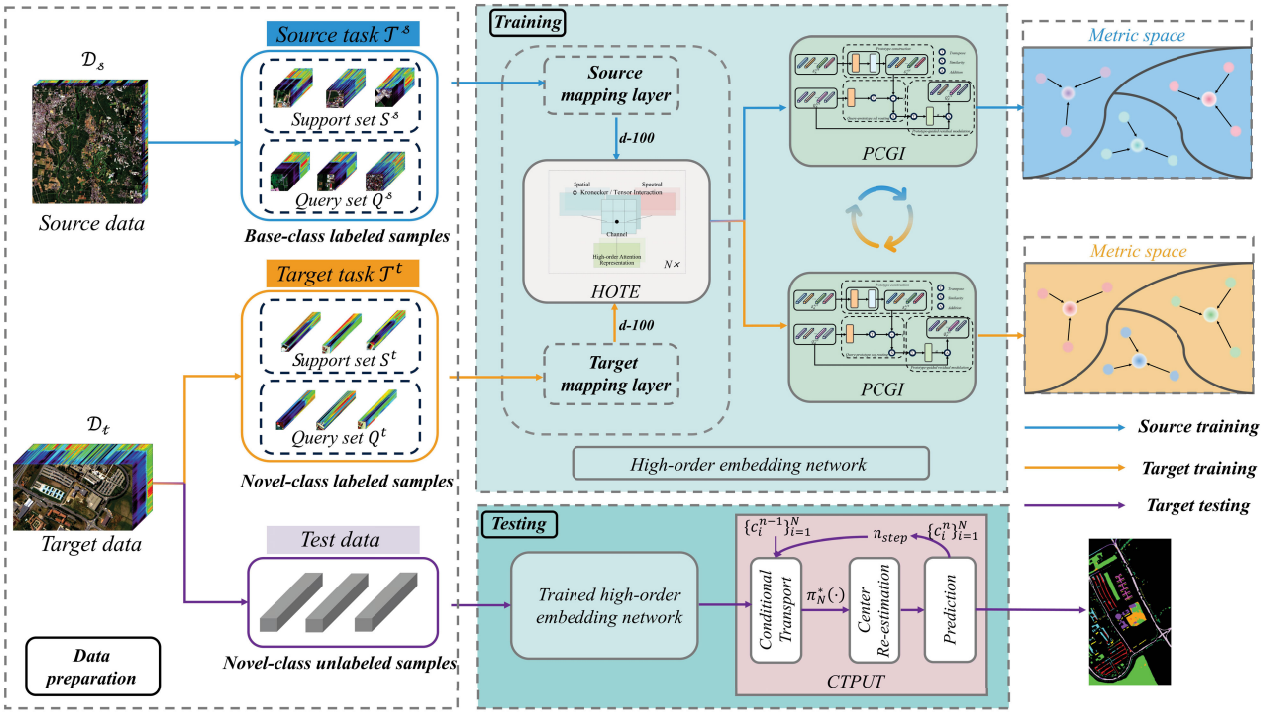


Fig. 1. Architecture of the proposed HOT-FSL.

To address the aforementioned challenges, we propose a TFSL framework with a high-order representation for cross-domain HSIC. The proposed framework is designed to jointly tackle three key issues: high-order spectral–spatial feature modeling, effective utilization of limited support information, and robustness under class-imbalanced query distributions. Specifically, we develop a dedicated high-order encoder to learn discriminative spectral–spatial representations tailored to hyperspectral data. We further introduce a prototype-guided interaction mechanism that strengthens the coupling between support and query features in few-shot scenarios. In addition, a conditional transport (CT)-based refinement strategy is incorporated to reduce prediction bias caused by class imbalance and to stabilize transductive inference.

The main contributions of the proposed framework are summarized as follows.

1) We propose a high-order transformer encoder (HOTE) that leverages Kronecker-decomposed multimodal attention and directional spatial modeling to efficiently capture high-order spatial–spectral dependencies in HSIs.
2) A prototype-conditioned gated interaction (PCGI) module injects support-set priors into query features, improving discriminability under cross-domain few-shot settings.
3) We propose conditional transport-based prototype-oriented unbiased transfer (CTPUT), a transductive prototype refinement strategy that adaptively calibrates class prototypes using query statistics, effectively mitigating bias caused by class imbalance.
4) A series of comparative experiments has demonstrated the superior performance of the proposed method.

The remainder of this article is structured as follows. Section II provides a detailed description of the proposed method. Section III presents extensive experiments and offers an in-depth analysis of the classification results. Finally, Section IV concludes this article.

II. Method

The overall architecture of the proposed high-order transformer-few shot learning (HOT-FSL) is illustrated in Fig. 1. We begin by presenting the problem formulation and data preparation for CDFSL to facilitate a clear understanding of the proposed methodology. Next, the three core modules of HOT-FSL are introduced in sequence, namely, HOTE, PCGI, and CTPUT, among which HOTE and PCGI jointly constitute a high-order feature encoding network.

A. Problem Formulation and Data Preparation

In CDFSL, two datasets are involved: a source-domain dataset $\mathcal{D}^s$ with abundant labeled samples and a target-domain dataset $\mathcal{D}^t$ with limited labeled samples. The source dataset $\mathcal{D}^s$ contains C_s classes, while the target dataset $\mathcal{D}^t$ consists of C_t classes, where $C_s \geq C_t$ generally holds. Based on these datasets, task sets $\mathcal{T}^s$ and $\mathcal{T}^t$ are constructed for the source and target domains, respectively. Each task $\mathcal{T}^{s/t}$ follows the N-way K-shot paradigm, where $N = C_t$ denotes the number of categories involved in each task and K represents the number of labeled samples available per category. Each task $\mathcal{T}^{s/t}$ is composed of a support set $\mathcal{S}^{s/t}$ and a query set $\mathcal{Q}^{s/t}$. The support set is represented as a 4-D tensor $\mathcal{S}^{s/t} \in \mathbb{R}^{(N\times K)\times S\times H\times W}$, while the query set is given by $\mathcal{Q}^{s/t} \in \mathbb{R}^{N_q\times S\times H\times W}$, where

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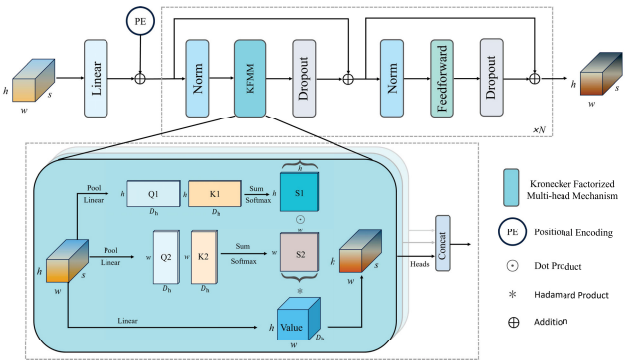


Fig. 2. Architecture of the proposed HOTE.

Q denotes the number of query samples per class. Here, S indicates the spectral dimensionality of samples from different target domains, and H and W correspond to the height and width of the extracted image patches, respectively. In other words, the task $\mathcal{T}^{s/t}$ serves as the input to the high-order embedding network and is jointly composed of the support set $\mathcal{S}^{s/t}$ and the query set $\mathcal{Q}^{s/t}$. During training, we follow the metalearning paradigm and adopt an alternating training strategy over the source-domain tasks $\mathcal{T}^s$ and the target-domain tasks $\mathcal{T}^t$. Under this setting, the model is required to leverage transferable knowledge learned from $\mathcal{D}^s$ to facilitate effective adaptation on $\mathcal{D}^t$ despite severe label scarcity.

B. HOTE

To comprehensively model the inherent high-order spatial–spectral dependencies in HSI, we design an HOTE, which explicitly captures multidimensional interactions between spatial locations and spectral channels. It should be noted that the term "high-order" does not refer to an explicit polynomial-order expansion but rather to the implicit modeling of complex nonlinear spatial–spectral dependencies through multiplicative feature interactions. Unlike conventional attention mechanisms that flatten features into sequences or mainly rely on first-order additive responses, HOTE preserves the 2-D spatial structure while enabling cross-dimensional multiplicative coupling between spatial and spectral representations. Such structured interaction modeling enhances the representation complexity of the learned features, thereby allowing the network to capture more discriminative nonlinear spatial–spectral correlations.

First, due to the pronounced spectral discrepancy between the source and target domains, dedicated mapping layers are introduced for the source-domain tasks $\mathcal{T}^s$ and the target-domain tasks $\mathcal{T}^t$, respectively, in order to facilitate subsequent feature modeling. Specifically, these mapping layers are designed to project the spectral dimensionality of tasks from different domains into a unified space with a fixed spectral dimension of 100. Each mapping layer is composed of a 2-D convolution followed by batch normalization; the obtained result is marked as $\mathcal{T}^{s/t}$ ($\mathcal{S}_a^{s/t}$, $\mathcal{Q}_a^{s/t}$). Next, these tensors will be encoded using the proposed HOTE. Fig. 2 shows the architecture of the proposed HOTE.

Without loss of generality, $\mathbf{X}$ is assumed to be the input to the proposed module, where $\mathbf{X}$ denotes the task-specific data associated with $\mathcal{T}^{s/t}$ and $\mathbf{X} \in \mathbb{R}^{b\times s\times h\times w}$. First, a spatial-resolution-preserving 1×1 2-D convolution is applied, which strengthens nonlinear feature representation and facilitates effective interactions across channels while maintaining the original feature map size. To embed absolute spatial positional information, a 2-D sinusoidal positional encoding is added to the feature representation, thereby supplying spatial priors for attention modeling while avoiding the introduction of extra learnable parameters, and the obtained feature map is denoted as X_{in}. Subsequently, the enhanced features are fed into a network composed of N Transformer-style blocks. Each block consists of layer normalization, a Kronecker-decomposition-based multimodal attention mechanism, dropout, and a feedforward network. The feedforward network is implemented using linear layers followed by the sigmoid-weighted linear unit (SiLU) activation function. In addition, residual connections are incorporated within each block to stabilize the training of deep networks and facilitate effective gradient propagation. Among these components, the Kronecker-decomposition-based attention mechanism serves as the core of the proposed HOTE. Specifically, attention is independently computed along different spatial dimensions, namely, the heightwise and widthwise directions, and the resulting attention responses are subsequently fused via summation. This design enables an efficient approximation of high-order multidimensional interactions while maintaining computational efficiency, thereby avoiding the prohibitive computational cost associated with full-tensor attention modeling. Section III-C provides a detailed description of the proposed Kronecker-decomposition-based attention mechanism.

Given the input feature tensor $\mathbf{X}_{in} \in \mathbb{R}^{B\times h\times w\times d}$, we first generate the query, key, and value representations via linear projections along the channel dimension

$$\mathcal{Q}^{(m)}, \mathcal{K}^{(m)} \in \mathbb{R}^{B\times h\times w\times d}, \quad \mathcal{V} \in \mathbb{R}^{B\times h\times w\times n_{\text{head}}\times d_h} \tag{1}$$

where $m \in \{1, 2\}$ denotes the spatial modes corresponding to the height and width dimensions, respectively, and $d_h = d/n_{\text{head}}$ is the dimensionality of each attention head. The query and key tensors are generated as mode-specific projections, while the value tensor is shared across different modes to preserve semantic consistency.

For each mode, attention is computed along a specific spatial dimension by reordering the tensor. Specifically, the query and key tensors are reshaped into multihead representations of size $\mathbb{R}^{B\times l\times n_{\text{head}}\times d_h}$, where $l \in \{h, w\}$ denotes the sequence length along the selected spatial axis. The attention for each mode is formulated as

$$\mathbf{S}^{(m)} = \text{Softmax}\left(\frac{\mathcal{Q}^{(m)}\left(\mathcal{K}^{(m)}\right)^{\top}}{\sqrt{d_h}}\right)\mathcal{V} \tag{2}$$

where the dot-product attention is performed along the corresponding spatial dimension while sharing the same value tensor $\mathcal{V}$. This design enables efficient modeling of spatial dependencies along each axis without introducing redundant parameters.

Unlike conventional axial attention that aggregates directional features in an additive manner, we adopt a multiplicative fusion strategy to capture higher-order spatial interactions.

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The final output is obtained by sequentially applying attention across the two modes

$$\mathbf{S} = \mathbf{S}^{(1)} \odot \mathbf{S}^{(2)} \tag{3}$$

where $\odot$ denotes the elementwise multiplication. Compared to the additive formulation, which treats each spatial mode independently and combines them via linear superposition, the multiplicative strategy introduces explicit interactions between different spatial directions. In particular, the response at each spatial location is jointly modulated by both modes such that the contribution of one direction depends on the activation of the other. This interaction mechanism effectively captures cross-dimensional correlations, rather than independent directional effects. From this perspective, the proposed formulation goes beyond separable modeling and encodes high-order dependencies since the final representation is determined by the joint configuration of multiple spatial modes instead of their individual contributions. This multiplicative formulation can be interpreted as a Kronecker-style factorization of full 2-D attention, which provides a compact yet expressive approximation of global attention while preserving cross-mode interactions.

C. Prototype-Conditioned Gated Interaction (PCGI)

Although HOTE is capable of extracting high-order spatial–spectral representations, the learned embeddings of support and query samples are still obtained independently. Under cross-domain few-shot scenarios, such a decoupled encoding manner often leads to insufficient utilization of class priors, thereby limiting the discriminability of query features. To address this issue, we propose a PCGI module, which explicitly establishes structured interactions between the support set and the query set through prototype modeling and residual guidance. Fig. 3 illustrates the details of the PCGI module.

Given the HOTE-encoded task $\mathcal{T}^{s/t} = (\mathcal{S}_h^{s/t}, \mathcal{Q}_h^{s/t})$, where $\mathcal{S}_h^{s/t} \in \mathbb{R}^{(N\times K)\times C\times H\times W}$ and $\mathcal{Q}_h^{s/t} \in \mathbb{R}^{(N\times Q)\times C\times H\times W}$, PCGI refines the query representations via three steps, namely, prototype extraction, soft routing, and prototype-guided residual modulation.

1) Prototype Extraction: For each class i, the support features are first averaged over the K shots to obtain a class-level representation

$$\mathbf{F}_i^s = \frac{1}{K}\sum_{j=1}^{K} \mathbf{f}_{i,j}^s \tag{4}$$

where $\mathbf{f}_{i,j}^s \in \mathbb{R}^{C\times H\times W}$. To obtain compact semantic descriptors, global average pooling is applied

$$\mathbf{p}_i = \mathrm{GAP}\left(\mathbf{F}_i^s\right) \in \mathbb{R}^C. \tag{5}$$

Stacking all class prototypes yields $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_N]^\top \in \mathbb{R}^{N\times C}$.

2) Query-to-Prototype Soft Routing: For each query feature $\mathbf{f}_r^q$, a global descriptor is computed as

$$\mathbf{z}_r = \mathrm{GAP}\left(\mathbf{f}_r^q\right). \tag{6}$$

The similarity between query descriptors and class prototypes is then measured via the inner product

$$a_{r,i} = \frac{\exp\left(\tau\, \mathbf{z}_r^\top \mathbf{p}_i\right)}{\sum_{k=1}^{N} \exp\left(\tau\, \mathbf{z}_r^\top \mathbf{p}_k\right)} \tag{7}$$

where τ is a learnable scaling factor. The routing vector $\mathbf{a}_r = [a_{r,1}, \ldots, a_{r,N}]$ indicates the degree to which the query sample is associated with each prototype.

3) Prototype-Guided Residual Modulation: A prototype mixture conditioned on the routing weights is obtained as

$$\tilde{\mathbf{p}}_r = \sum_{i=1}^{N} a_{r,i}\, \mathbf{p}_i. \tag{8}$$

The mixed prototype is subsequently projected and injected into the query feature through a residual pathway

$$\hat{\mathbf{f}}_r^q = \mathbf{f}_r^q + \alpha\, \psi\left(\tilde{\mathbf{p}}_r\right) \tag{9}$$

where $\psi(\cdot)$ denotes a 1×1 convolution and α is a learnable coefficient.

Through this design, query features are adaptively modulated according to class-aware priors inferred from the support set, which effectively enhances interclass separability while maintaining intraclass consistency. It is worth emphasizing that PCGI introduces no manually tuned hyperparameters, and all transformations are optimized jointly in an end-to-end manner. Through the modeling of HOTE, $\mathcal{T}^{s/t}$ $(\mathcal{S}_h^{s/t}, \mathcal{Q}_h^{s/t})$ is ultimately expressed as $\mathcal{T}^{s/t}$ $(\mathcal{S}_p^{s/t}, \mathcal{Q}_p^{s/t})$, which enables effective integration of support set and query set representations, and provides reliable embeddings for the subsequent transductive refinement.

D. CTPUT

During inference, owing to the inherent correlation between the support and query sets, class prototypes initialized from the support set are expected to be further refined by exploiting statistical information contained in the query samples. However, how to reliably and unbiasedly transfer such statistics for prototype calibration remains a nontrivial problem. To address this issue, a CTPUT mechanism is proposed, in which discriminative statistical cues extracted from the query set are explicitly leveraged to progressively reduce the discrepancy between the estimated prototypes and the underlying true class prototypes. In the following, the theoretical foundation of CT is first reviewed, followed by a CT-based unbiased development strategy and a transductive inference mechanism tailored for few-shot classification.

1) CT Theory: CT provides a principled framework for measuring discrepancies between general probability distributions and has demonstrated strong empirical effectiveness in various learning scenarios. Let $p_X(x)$ and $p_Y(y)$ denote two discrete distributions defined on spaces $\mathcal{S}_p^{s/t}$ and $\mathcal{Q}_p^{s/t}$, respectively. A transport plan $\pi(x, y)$ is constructed to couple these two distributions. In the forward CT formulation, the joint distribution is factorized as

$$\pi(x, y) = p_X(x)\, \pi_Y(y \mid x) \tag{10}$$

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where $\pi_Y(y \mid x)$ denotes the conditional distribution of y given x. By construction, the marginal constraint $\int \pi(x,y)\,dy = p_X(x)$ is satisfied, whereas $\int \pi(x,y)\,dx \neq p_Y(y)$ generally holds.

Let $d_\phi(h_1, h_2) \in \mathbb{R}$ denote a learnable discrepancy function defined as

$$d_\phi(h_1, h_2) = -g_\phi((h_1 - h_2) \odot (h_1 - h_2)) \tag{11}$$

where $\odot$ denotes the elementwise multiplication and g_ϕ is a multilayer perceptron parameterized by ϕ. To incorporate the target distribution $p_Y(y)$ as a prior while relaxing the marginal constraint on Y, the exponential term $e^{-d_\phi(x,y)}$ is interpreted as an unnormalized likelihood. According to Bayes' rule, the forward conditional distribution is defined as

$$\pi_Y(y \mid x) = \frac{e^{-d_\phi(x,y)}\, p_Y(y)}{Q(x)}$$
$$Q(x) = \int e^{-d_\phi(x,y)}\, p_Y(y)\, dy \tag{12}$$

where $Q(x)$ is a normalization constant ensuring $\int \pi_Y(y \mid x)\,dy = 1$. This conditional distribution is referred to as the forward *navigator*. Based on this formulation, the forward CT cost is defined as

$$\mathcal{C}(X \to Y) = \mathbb{E}_{x\sim p_X(x)} \mathbb{E}_{y\sim \pi_Y(\cdot|x)} \left[c(x,y)\right]. \tag{13}$$

By reversing the transport direction, the backward CT is constructed by factorizing the joint distribution as

$$\pi(x,y) = p_Y(y)\, \pi_X(x \mid y) \tag{14}$$

where the backward navigator is given by

$$\pi_X(x \mid y) = \frac{e^{-d_\phi(x,y)}\, p_X(x)}{Q(y)}$$
$$Q(y) = \int e^{-d_\phi(x,y)}\, p_X(x)\, dx. \tag{15}$$

The corresponding backward CT cost is defined as

$$\mathcal{C}(X \leftarrow Y) = \mathbb{E}_{y\sim p_Y(y)} \mathbb{E}_{x\sim \pi_X(\cdot|y)} \left[c(x,y)\right]. \tag{16}$$

By combining the forward and backward formulations, the overall CT cost is defined as

$$\mathcal{C}_\rho(X,Y) = \rho\, \mathcal{C}(X \to Y) + (1-\rho)\, \mathcal{C}(X \leftarrow Y) \tag{17}$$

where $\rho \in [0,1]$ controls the tradeoff between mode-covering and mode-seeking behaviors.

2) CT-Based Algorithm for Unbiased Development: The initial class prototypes $\{\mathbf{c}_i^{(0)}\}_{i=1}^N$ are computed by averaging the feature representations of the support samples for each class

$$\mathbf{c}_i^{(0)} = \frac{1}{K} \sum_{j=1}^{K} \mathbf{f}_{i,j}^s, \quad i = 1, \ldots, N \tag{18}$$

where $\mathbf{f}_{i,j}^s$ denotes the feature of the jth support sample from class i.

Meanwhile, all query samples are projected into the same feature space, yielding representations $\{\mathbf{f}_r^q\}_{r=1}^M$. By treating the prototypes and query features as samples drawn from two discrete distributions, CT is employed to learn the corresponding transport matrices by minimizing the CT objective in (17). This process enables the extraction of unbiased statistical information from the query set.

Algorithm 1 CTPUT

Input: Support set $\{\mathbf{x}_{i,j}^s\}_{j=1}^K$, $i = 1, \ldots, N$; Query set $\{\mathbf{x}_r^q\}_{r=1}^M$
Feature Extraction: Extract support features $\mathbf{f}_{i,j}^s$ and query features $\mathbf{f}_r^q$ using the trained encoder
Prototype Initialization: $\mathbf{c}_i^{(0)} = \frac{1}{K}\sum_{j=1}^K \mathbf{f}_{i,j}^s$, $i = 1, \ldots, N$
for $n = 0$ to $n_{\text{step}} - 1$ **do**
  E-step (CT Optimization): Initialize navigator parameters ϕ
  Optimize ϕ^* by minimizing CT cost in Eq. (17) with fixed prototypes $\{\mathbf{c}_i^{(n)}\}$
  Compute forward transport matrix $\hat{\pi}_N^*$ via Eq. (15)
  M-step (Prototype Refinement): Compute refined prototypes $\mathbf{c}_i'$ using Eq. (10)
  Update prototypes with inertia: $\mathbf{c}_i^{(n+1)} = \mathbf{c}_i^{(n)} + \alpha(\mathbf{c}_i' - \mathbf{c}_i^{(n)})$
end for
Prediction: Assign label for each query sample according to Eq. (13)
Output: Refined prototypes $\{\mathbf{c}_i^{(n_{\text{step}})}\}$ and predictions

3) Transductive Inference Mechanism for Unbiased Transfer: Once the optimal forward transport matrix $\hat{\pi}_N^*$ is obtained by minimizing (17), refined class prototypes are computed as

$$\mathbf{c}_i' = \frac{\sum_{r=1}^M \hat{\pi}_{N,r,i}^* \mathbf{f}_r^q + \sum_{j=1}^K \mathbf{f}_{i,j}^s}{K + \sum_{r=1}^M \hat{\pi}_{N,r,i}^*}, \quad i = 1, \ldots, N. \tag{19}$$

Once the navigator parameters ϕ^* are fixed, the class prototypes constitute the only remaining variables to be optimized. Minimizing the CT objective with respect to the prototypes yields the closed-form solution in (19). To stabilize the refinement process, an inertia-based update strategy is adopted

$$\mathbf{c}_i^{(n+1)} = \mathbf{c}_i^{(n)} + \alpha\left(\mathbf{c}_i' - \mathbf{c}_i^{(n)}\right), \quad i = 1, \ldots, N \tag{20}$$

where $\alpha \in (0,1)$ controls the update rate.

The estimation of the transport matrix and the refinement of prototypes are inherently coupled, while the transport matrix is stochastic, and the prototypes are deterministic. These properties naturally motivate an expectation–maximization (EM) optimization scheme.

1) *E-Step:* Given fixed prototypes, we estimate the transport matrix $\hat{\pi}_N^*$ by optimizing the CT objective. Specifically, the discrepancy function $d_\phi(\cdot,\cdot)$ is instantiated via a parametric navigator network, whose parameters ϕ are updated using gradient-based optimization (e.g., backpropagation) to minimize the transport loss. The optimized d_ϕ then induces the transport matrix via a softmax operation.
2) *M-Step:* Given the estimated transport matrix $\hat{\pi}_N^*$, we update the prototypes using (20), where the update follows an inertia-controlled formulation.

In practice, the navigator parameters ϕ are optimized for multiple inner iterations within each E-step using stochastic gradient descent. To improve stability and avoid bias accumulation across EM iterations, the navigator is reinitialized at the beginning of each E-step. The complete procedure is summarized in Algorithm 1.

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The estimation of the transport matrix and the refinement of prototypes are inherently coupled, while the transport matrix is stochastic, and the prototypes are deterministic. These properties naturally motivate an EM optimization scheme.

1) *E-Step:* Update $\hat{\pi}_N^*$ by minimizing the CT objective with fixed prototypes.
2) *M-Step:* Update the prototypes using (20) with fixed $\hat{\pi}_N^*$.

At each iteration, the navigator parameters are reinitialized to ensure stable convergence. The complete procedure is summarized in Algorithm 1.

After n_{step} EM iterations, the final prediction for each query sample is obtained as

$$y_r^q = \arg\max_{i\in\{1,\ldots,N\}} \hat{\pi}_N\left(\mathbf{c}_i^{(n_{\text{step}})} \mid \mathbf{f}_r^q, \phi^*\right), \quad r = 1,\ldots,M. \tag{21}$$

The overall performance is evaluated by averaging the classification accuracy over all query samples across multiple few-shot tasks.

III. Experimental Results and Analysis

In this section, we first introduce the datasets used in the experiments along with the corresponding experimental settings. Subsequently, extensive comparative experiments and ablation studies are conducted to comprehensively validate the effectiveness of the proposed HOT-FSL framework. Finally, a complexity analysis is carried out to evaluate the computational efficiency and practical applicability of the proposed method in comparison with existing approaches.

A. Dataset Description

The experimental datasets are partitioned into two subsets: a source dataset containing base classes with sufficient labeled data and target datasets corresponding to novel classes with few-shot annotations. Following [29], [32], and [49], the Chikusei dataset is selected as the source dataset, while the Pavia University (PU), Indian Pines (IP), Houston2013 (HU), and Salinas Valley (SV) datasets are used as target datasets.

1) *Chikusei:* [49] The Chikusei dataset was collected in 2014 over Chikusei, Japan, using the hyperspectral visible/near-infrared camera (Hyperspec-VNIR-C). The dataset has a spatial size of 2517 × 2335 pixels and includes 19 annotated land-cover classes. The spatial resolution is 2.5 m per pixel, and the hyperspectral data consist of 128 spectral bands spanning wavelengths from 363 to 1018 nm.
2) *PU:* The PU dataset was acquired by the reflective optics system imaging spectrometer (ROSIS) during a flight campaign over Pavia, Northern Italy. The dataset has a spatial size of 610 × 340 pixels and consists of 103 spectral bands covering wavelengths from 0.4 to 0.86 μm. It contains 10 249 labeled samples distributed across nine land-cover classes.
3) *IP:* The IP dataset was acquired in 1992 over Northwestern Indiana, USA, using the airborne visible/infrared imaging spectrometer (AVIRIS). The scene has a spatial size of 145 × 145 pixels and contains 200 spectral bands covering wavelengths from 0.4 to 2.5 μm. A total of 16 ground-cover classes are annotated in this dataset.
4) *HU:* The HU dataset was jointly provided by the HSI Analysis Group at the University of Houston and the National Center for Airborne Laser Mapping, with support from the National Science Foundation. The dataset consists of 349 × 1905 pixels and contains 144 spectral bands ranging from 0.38 to 1.05 μm, covering a total of 15 land-cover categories.
5) *SV:* The SV dataset was collected by the AVIRIS sensor over SV, California, in 1998. After removing 20 noisy and water-absorption bands, 204 effective spectral bands are retained for HSIC. The dataset has a spatial size of 512 × 217 pixels and includes 16 land-cover classes.

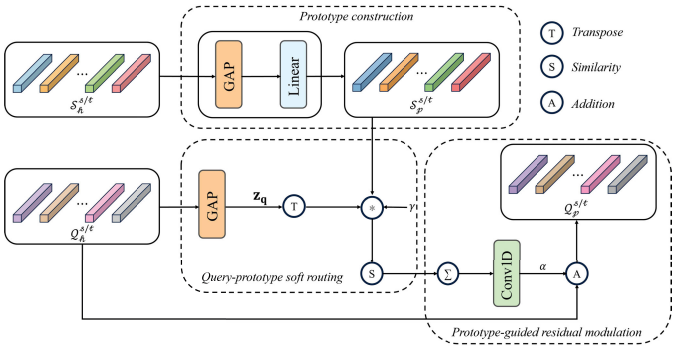


Fig. 3. Architecture of the proposed PCGI.

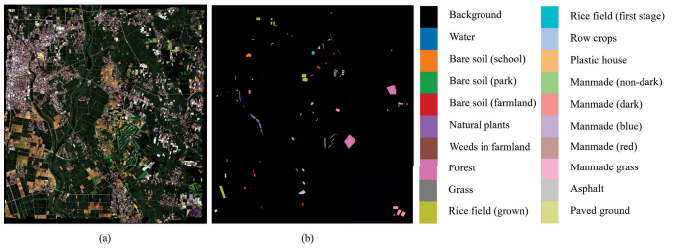


Fig. 4. Overview of the Chikusei dataset. (a) Pseudocolor image. (b) Ground-truth map.

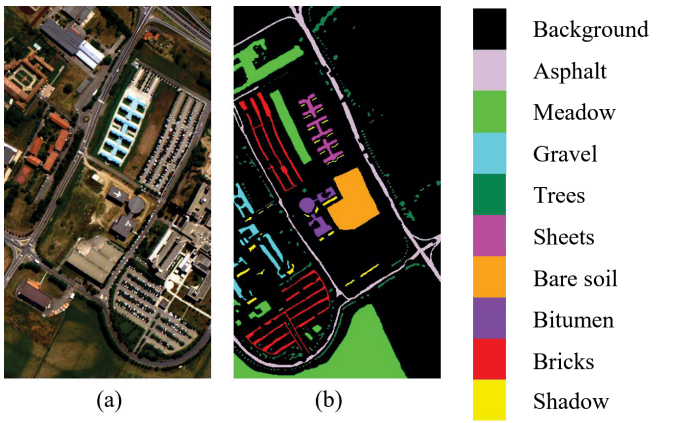


Fig. 5. Overview of the PU dataset. (a) Pseudocolor image. (b) Ground-truth map.

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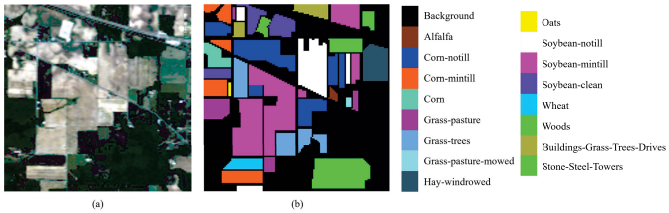


Fig. 6. Overview of the IP dataset. (a) Pseudocolor image. (b) Ground-truth map.

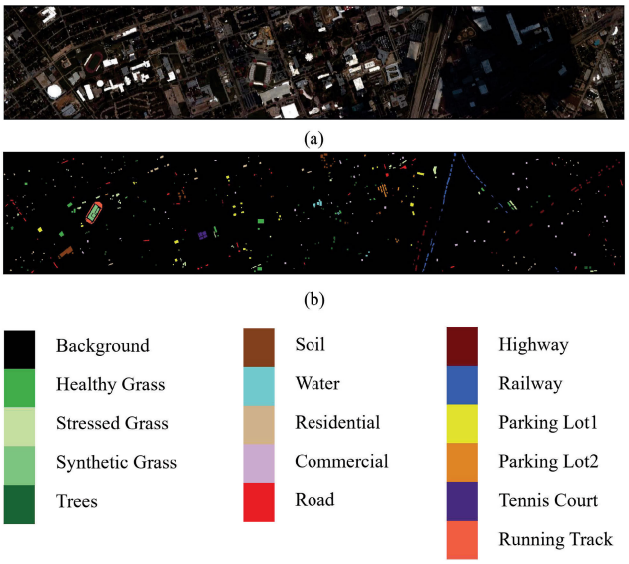


Fig. 7. Overview of the HU dataset. (a) Pseudocolor image. (b) Ground-truth map.

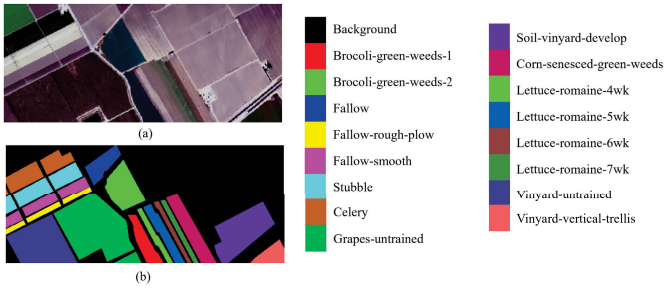


Fig. 8. Overview of the SV dataset. (a) Pseudocolor image. (b) Ground-truth map.

Table I summarizes the land-cover class information of the source-domain dataset, whereas Table II reports the corresponding details for the target-domain datasets. Fig. 4 illustrates the source-domain dataset, while Figs. 5–8 provide visualizations of the target-domain datasets. These visual representations facilitate a clearer and more intuitive understanding of the datasets.

B. Experimental Settings and Evaluation Metrics

To ensure the effectiveness of the proposed networks, comprehensive experimental settings are adopted. During the metatraining stage, minibatches (i.e., episodes) constructed from the source and target datasets are alternately fed into the network. The total number of training episodes is empirically set to 10 000, which is sufficient to guarantee stable convergence. The learning rate is fixed at 0.0005 for all datasets. In the source domain, the number of support and query samples is set to 1 and 19, respectively. For all experiments except the sensitivity analysis, the target domain uses five support samples and 15 query samples per class. The spatial patch size in each episode is set to 9×9, following the common practice in recent few-shot HSIC studies. In addition, the number of sampled classes is configured as nine for the UP dataset, 16 for the IP and SV datasets, and 15 for the HU dataset, respectively. This configuration is designed to promote the collaborative learning of a shared discriminative embedding space across both source and target classes.

The HOT-FSL has been implemented using Python and the PyTorch framework, and all subsequent experiments have been implemented on the same hardware platform with a GTX-4090 GPU, Intel 4210R CPU, and 45-GB memory. The classification performance is evaluated using the overall accuracy (OA), average accuracy (AA), and the kappa coefficient. To ensure statistical reliability and avoid biased evaluations, all experiments are conducted over ten independent runs.

TABLE I
LAND-COVER CLASSES AND SAMPLE COUNTS IN THE CHIKUSEI DATASET

Class ID	Land cover type	Samples
C01	Water	2,345
C02	Bare soil (school)	2,859
C03	Bare soil (park)	236
C04	Bare soil (farmland)	48,525
C05	Natural plants	4,297
C06	Weeds in farmland	1,108
C07	Forest	20,516
C08	Grass	6,515
C09	Rice field (grown)	13,369
C10	Rice field (first stage)	1,268
C11	Row crops	5,961
C12	Plastic house	2,193
C13	Manmade (non-dark)	1,220
C14	Manmade (dark)	7,664
C15	Manmade (blue)	431
C16	Manmade (red)	222
C17	Manmade grass	1,040
C18	Asphalt	801
C19	Paved ground	145
Total		**77,592**

C. Sensitivity Analysis of Hyperparameter

We evaluate the sensitivity of HOTE with respect to the number of Transformer blocks N and the number of attention heads h on four benchmark datasets, including PU, IP, HU, and SV. The influence of these hyperparameters is visually illustrated in Fig. 9. Overall, the proposed model demonstrates stable performance under different configurations, with accuracy fluctuations generally within a narrow range. Specifically, moderate depth consistently yields strong results across datasets. For instance, the best performance is typically achieved when $N = 2$, while increasing the depth to

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TABLE II
LAND-COVER CLASSES AND SAMPLE COUNTS FOR THE PU, IP, HU, AND SV DATASETS

Class	PU Dataset		IP Dataset		HU Dataset		SV Dataset	
	Type	Samples	Type	Samples	Type	Samples	Type	Samples
C01	Asphalt	6,631	Alfalfa	46	Healthy grass	1,251	Brocoli_green_weeds_1	2,009
C02	Meadow	18,549	Corn-notill	1,428	Stressed grass	1,254	Brocoli_green_weeds_2	3,726
C03	Gravel	2,099	Corn-mintill	830	Synthetic grass	697	Fallow	1,976
C04	Trees	3,064	Corn	237	Healthy grass	1,244	Fallow_rough_plow	1,394
C05	Sheets	1,345	Grass-pasture	483	Soil	1,242	Fallow_smooth	2,678
C06	Bare soil	5,029	Grass-trees	730	Water	325	Stubble	3,959
C07	Bitumen	1,330	Grass-pasture-mowed	28	Residential	1,268	Celery	379
C08	Bricks	3,682	Hay-windrowed	478	Commercial	1,244	Grapes_untrained	11,271
C09	Shadow	947	Oats	20	Road	1,252	Soil_vinyard_develop	6,203
C10			Soybean-notill	972	Highway	1,227	Corn_senesced_green_weeds	3,278
C11			Soybean-mintill	2,455	Railway	1,235	Lettuce_romaine_4wk	1,068
C12			Soybean-clean	593	Parking lot1	1,233	Lettuce_romaine_5wk	1,927
C13			Wheat	205	Parking lot2	469	Lettuce_romaine_6wk	916
C14			Woods	1,265	Tennis court	428	Lettuce_romaine_7wk	1,070
C15			Buildings-Grass-Trees-Drives	386	Running track	660	Vinyard_untrained	7,268
C16			Stone-Steel-Towers	93			Vinyard_vertical_trellis	1,807
Total		**42,776**	-	**10,249**	-	**15,202**	-	**54,129**

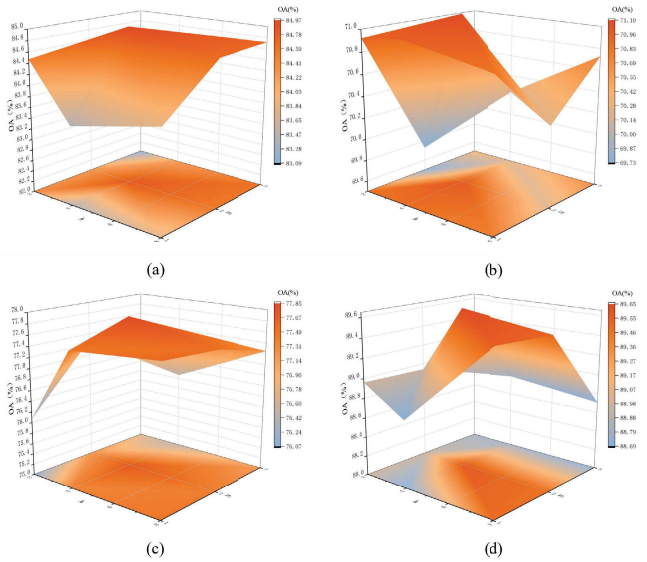


Fig. 9. Sensitivity of the proposed model to the number of Transformer blocks N and attention heads h on four benchmark datasets. (a) PU. (b) IP. (c) HU. (d) SV.

three blocks brings marginal improvement or slight degradation, especially on IP and PU. This phenomenon suggests that excessively deep modeling may introduce redundant capacity and lead to overfitting in few-shot scenarios. Regarding the number of attention heads, the performance variations among $h = \{2, 4, 8\}$ remain limited on all datasets. The accuracy differences are mostly within about one percentage point, indicating that the Kronecker-decomposed directional attention is not highly sensitive to the head number and can maintain reliable representation ability even with a compact configuration. These observations verify that HOTE achieves robust performance without relying on delicate hyperparameter tuning.

In addition, Fig. 10 presents the sensitivity analysis with respect to the tradeoff parameter ρ, the inertia coefficient α, and the number of EM iterations n_{step}, respectively. Specifically, the parameter ρ controls the balance between forward and backward transport in the proposed CT mechanism, thereby influencing the reliability of the soft assignment. As observed, the performance consistently improves as ρ increases from 0.1 to 0.5, achieving the best results around $\rho = 0.5$. When ρ becomes larger, a slight performance degradation can be observed, indicating that overemphasizing either direction may weaken the complementary effect of bidirectional matching. The inertia coefficient α determines the update magnitude of class prototypes during the EM-like refinement, which directly affects convergence stability. As shown in the figure, moderate values lead to the most favorable performance across datasets. In contrast, overly small α slows down the adaptation process, while excessively large values may cause unstable updates and degrade performance. The parameter n_{step} denotes the number of EM iterations for transductive refinement. Increasing n_{step} from 5 to 20 significantly improves performance, demonstrating the effectiveness of iterative prototype refinement. However, further increasing the number of iterations brings limited gains or slight declines, suggesting that excessive iterations may introduce noise accumulation or overfitting to unreliable pseudolabels. Overall, these results indicate that the proposed method achieves optimal performance under moderate parameter settings and maintains stable behavior across a relatively wide range of configurations, highlighting its robustness in practical applications.

D. Comparative Experiments

To validate the effectiveness of the proposed HOT-FSL, its classification performance is systematically compared with that of several state-of-the-art methods, with eight advanced algorithms selected as baseline approaches for comparison. Specifically, GiaFSL [45] and ADAFSL [40] are representative early studies that introduce the CDFSL paradigm into HSIC. Moreover, ECSA [50], GCC-FSL [44], MLPA [51], CTF-SSCL [52], and FDFSL [53] are recently proposed methods

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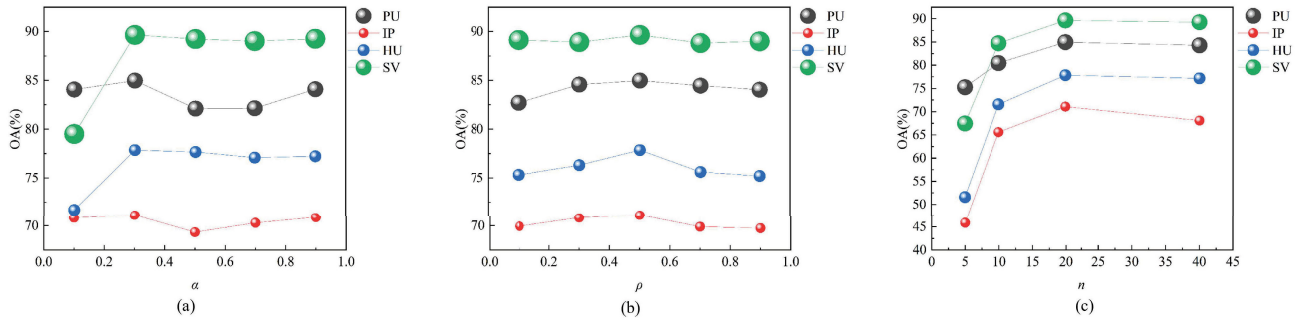


Fig. 10. Sensitivity analysis of key hyperparameters, including ρ (balancing forward and backward transport in CT), α (controlling the update magnitude of class prototypes), and n_{step} (denoting the number of EM iterations in transductive refinement). (a) Denotes α, (b) denotes ρ, and (c) denotes n_{step}.

TABLE III
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE PU DATASET

No.	GiaFSL	ECSA	GCC-FSL	ADAFSL	MLPA	TEFSL	CTF-SSCL	FDFSL	HOT-FSL
1	75.86 ± 9.20	88.72 ± 0.95	77.65 ± 3.07	82.87 ± 1.99	88.79 ± 4.25	60.78 ± 8.64	**90.99 ± 5.78**	80.59 ± 4.82	81.23 ± 5.79
2	78.68 ± 9.51	76.15 ± 3.18	86.40 ± 5.20	68.59 ± 14.58	83.15 ± 2.39	91.87 ± 3.63	82.56 ± 3.07	**94.71 ± 0.82**	92.33 ± 6.10
3	62.56 ± 13.01	61.23 ± 2.41	69.47 ± 4.93	70.18 ± 4.50	81.51 ± 6.10	66.79 ± 7.11	73.47 ± 4.51	53.41 ± 4.55	**96.30 ± 2.64**
4	86.83 ± 4.52	88.22 ± 1.67	**92.81 ± 3.13**	90.54 ± 5.65	84.58 ± 4.47	89.65 ± 5.38	88.64 ± 3.18	92.51 ± 1.22	72.86 ± 10.67
5	94.03 ± 1.58	**100.00 ± 0.00**	99.15 ± 1.04	99.23 ± 0.93	95.99 ± 2.22	98.93 ± 0.99	99.45 ± 0.32	98.19 ± 0.64	99.94 ± 0.06
6	58.68 ± 4.25	64.94 ± 2.12	56.28 ± 7.65	75.73 ± 11.56	73.65 ± 6.98	51.45 ± 14.92	**78.22 ± 2.01**	56.48 ± 3.91	72.77 ± 13.94
7	76.21 ± 6.45	77.93 ± 1.35	81.39 ± 4.61	90.44 ± 4.05	80.59 ± 2.69	89.07 ± 5.42	97.71 ± 1.50	76.97 ± 3.67	**96.79 ± 3.29**
8	67.29 ± 7.54	78.43 ± 2.75	**79.91 ± 4.22**	77.55 ± 8.98	55.58 ± 6.48	58.80 ± 15.03	55.91 ± 14.38	78.60 ± 4.30	63.64 ± 8.36
9	98.49 ± 1.50	97.94 ± 0.38	98.62 ± 1.53	99.40 ± 0.28	98.98 ± 0.62	**99.62 ± 0.29**	98.53 ± 1.26	98.13 ± 0.98	90.32 ± 7.37
OA	75.55 ± 3.80	78.39 ± 1.38	81.09 ± 2.24	76.38 ± 5.37	81.23 ± 0.82	78.37 ± 2.29	82.40 ± 3.46	84.09 ± 1.28	**84.97 ± 1.73**
AA	77.63 ± 1.38	81.51 ± 0.34	82.41 ± 1.15	83.84 ± 1.78	82.54 ± 0.92	78.55 ± 3.17	85.05 ± 2.67	81.07 ± 1.35	**85.13 ± 2.27**
Kappa	68.59 ± 4.27	72.09 ± 1.63	75.22 ± 2.65	70.38 ± 5.88	75.68 ± 0.93	71.48 ± 3.22	77.32 ± 4.32	78.70 ± 1.73	**80.13 ± 2.08**

TABLE IV
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE IP DATASET

No.	GiaFSL	ECSA	GCC-FSL	ADAFSL	MLPA	TEFSL	CTF-SSCL	FDFSL	HOT-FSL
1	84.88 ± 12.47	98.05 ± 1.83	**100.00 ± 0.00**	98.37 ± 2.30	86.34 ± 14.02	99.51 ± 0.98	**97.56 ± 0.00**	99.19 ± 1.15	**100.00 ± 0.00**
2	31.72 ± 5.63	39.49 ± 7.82	48.18 ± 4.14	35.75 ± 4.69	42.54 ± 8.21	44.36 ± 8.69	41.18 ± 7.07	54.81 ± 8.28	**63.71 ± 8.89**
3	41.26 ± 5.67	46.30 ± 8.88	47.84 ± 5.01	58.79 ± 5.76	53.09 ± 10.65	53.38 ± 5.21	52.85 ± 6.43	**63.88 ± 1.69**	63.76 ± 6.29
4	78.88 ± 3.39	75.78 ± 7.54	67.67 ± 3.71	92.24 ± 1.53	86.72 ± 4.13	**93.11 ± 3.89**	80.60 ± 2.20	72.41 ± 4.00	89.83 ± 5.38
5	63.47 ± 9.16	66.28 ± 9.45	82.01 ± 3.65	83.61 ± 1.58	71.92 ± 6.98	66.82 ± 6.53	**85.78 ± 2.67**	82.08 ± 2.94	82.97 ± 1.46
6	83.26 ± 4.84	76.39 ± 10.73	79.59 ± 4.91	85.38 ± 7.73	80.41 ± 9.16	80.72 ± 7.85	84.99 ± 6.36	75.03 ± 5.02	**94.68 ± 3.23**
7	**100.00 ± 0.00**	**100.00 ± 0.00**	98.55 ± 2.05	**100.00 ± 0.00**	99.13 ± 1.74	99.13 ± 1.74	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
8	92.81 ± 3.65	70.36 ± 14.45	79.14 ± 6.49	95.14 ± 1.75	89.51 ± 6.00	73.32 ± 9.37	97.72 ± 2.12	84.36 ± 5.38	**99.92 ± 0.10**
9	**100.00 ± 0.00**	98.67 ± 2.67	**100.00 ± 0.00**	**100.00 ± 0.00**	98.67 ± 2.67	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	98.67 ± 2.67
10	62.38 ± 3.48	56.90 ± 8.60	59.50 ± 4.05	65.56 ± 1.83	67.22 ± 3.91	69.04 ± 2.38	70.78 ± 4.32	57.53 ± 4.27	**72.97 ± 0.79**
11	64.20 ± 4.73	46.08 ± 10.13	62.78 ± 4.86	59.29 ± 13.21	56.09 ± 17.00	63.58 ± 4.58	65.42 ± 3.65	**67.27 ± 8.70**	50.92 ± 8.01
12	38.71 ± 5.35	51.53 ± 10.22	47.05 ± 5.50	46.71 ± 10.00	45.71 ± 11.32	41.84 ± 1.81	30.44 ± 2.39	53.68 ± 4.38	**56.06 ± 9.55**
13	97.80 ± 1.96	97.20 ± 1.54	97.83 ± 0.24	95.33 ± 5.91	96.60 ± 5.43	**100.00 ± 0.00**	99.10 ± 1.32	97.67 ± 1.55	99.70 ± 0.30
14	79.65 ± 1.95	83.37 ± 9.53	76.46 ± 4.43	84.95 ± 2.47	86.14 ± 5.50	95.18 ± 1.30	82.32 ± 5.85	**91.16 ± 4.57**	90.97 ± 4.47
15	57.38 ± 8.17	72.39 ± 8.61	57.83 ± 6.01	81.45 ± 12.74	78.64 ± 8.98	60.68 ± 4.73	**90.71 ± 3.41**	68.24 ± 4.73	62.52 ± 12.11
16	94.09 ± 1.51	97.95 ± 2.08	**100.00 ± 0.00**	98.11 ± 2.68	96.36 ± 6.20	99.77 ± 0.46	**100.00 ± 0.00**	98.48 ± 0.54	**100.00 ± 0.00**
OA	61.93 ± 1.77	58.91 ± 2.62	64.08 ± 2.22	66.61 ± 2.21	65.14 ± 4.20	66.94 ± 1.55	67.96 ± 1.50	69.82 ± 2.75	**71.10 ± 2.17**
AA	73.15 ± 1.57	73.55 ± 1.17	75.28 ± 0.23	80.04 ± 0.59	77.19 ± 2.16	77.53 ± 0.92	79.97 ± 0.80	79.11 ± 1.13	**82.91 ± 0.95**
Kappa	56.86 ± 1.94	54.13 ± 2.74	59.45 ± 2.37	62.40 ± 2.33	60.85 ± 4.56	62.66 ± 1.64	63.92 ± 1.58	65.77 ± 2.96	**67.69 ± 2.37**

that explicitly address the domain discrepancy issue in few-shot HSIC and have demonstrated competitive performance. TEFSL [48] is the first work to introduce TFSL into the field of HSIC, providing new possibilities for addressing such problems. To ensure a fair and consistent experimental comparison, the Chikusei dataset is uniformly selected as the source-domain dataset for all comparison methods. For the source domain, the number of support and query samples is fixed at 1 and 19, respectively, thereby guaranteeing identical FSL conditions across all algorithms. The number of categories is dynamically adjusted according to the characteristics of different target datasets. The key parameters in these methods are aligned with the settings reported in the original literature. Furthermore, to ensure the reliability of the experimental results, ten independent runs have been conducted for each method and dataset.

1) Quantitative Analysis: Tables III–VI present the quantitative performance comparison of all methods under the five-shot setting, where each class contains five labeled samples and the best results are highlighted in bold. Specifically,

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TABLE V
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE HU DATASET

No.	GiaFSL	ECSA	GCC-FSL	ADAFSL	MLPA	TEFSL	CTF-SSCL	FDFSL	HOT-FSL
1	98.25 ± 0.26	85.96 ± 4.33	**98.51 ± 0.99**	93.85 ± 6.38	85.36 ± 0.17	85.09 ± 0.75	89.99 ± 7.44	82.01 ± 1.35	96.04 ± 2.18
2	77.41 ± 3.20	77.87 ± 5.79	79.42 ± 1.92	73.63 ± 5.09	75.34 ± 4.48	59.63 ± 16.44	78.80 ± 12.01	**91.72 ± 2.03**	83.91 ± 3.03
3	92.17 ± 2.68	95.95 ± 3.57	94.42 ± 1.77	98.41 ± 2.05	99.51 ± 0.51	98.50 ± 0.48	98.27 ± 1.42	94.42 ± 3.82	**99.91 ± 0.07**
4	88.73 ± 1.25	89.78 ± 2.59	88.10 ± 2.33	89.16 ± 5.98	91.53 ± 1.04	92.74 ± 1.62	90.57 ± 3.69	**94.35 ± 0.65**	87.27 ± 8.08
5	93.47 ± 2.09	99.11 ± 0.75	93.35 ± 1.59	91.92 ± 3.98	95.25 ± 1.38	**99.56 ± 0.30**	92.77 ± 5.76	98.93 ± 0.75	98.25 ± 0.72
6	82.88 ± 2.46	**85.44 ± 1.21**	77.00 ± 15.96	83.85 ± 0.53	81.25 ± 1.93	81.25 ± 2.74	83.63 ± 6.60	78.94 ± 2.38	85.41 ± 0.15
7	62.90 ± 7.24	**76.33 ± 3.43**	64.62 ± 4.10	67.41 ± 7.64	72.57 ± 2.60	67.49 ± 2.67	71.54 ± 11.52	66.49 ± 4.54	51.15 ± 14.71
8	39.44 ± 1.10	34.08 ± 6.71	44.91 ± 7.41	38.77 ± 5.72	44.00 ± 7.21	**56.82 ± 1.38**	38.05 ± 10.82	47.67 ± 2.27	53.73 ± 3.95
9	**83.95 ± 2.04**	80.37 ± 2.72	83.83 ± 1.07	60.36 ± 11.27	60.85 ± 5.09	55.19 ± 2.24	77.02 ± 11.17	62.71 ± 3.99	53.25 ± 12.55
10	50.00 ± 3.43	67.38 ± 16.33	50.47 ± 2.74	**69.61 ± 11.32**	52.05 ± 7.22	56.79 ± 9.93	60.53 ± 9.46	66.60 ± 5.32	55.24 ± 22.73
11	61.02 ± 6.07	50.81 ± 17.33	61.30 ± 6.25	53.47 ± 6.27	61.77 ± 7.29	56.08 ± 3.82	64.11 ± 10.31	66.47 ± 4.78	**78.56 ± 16.07**
12	68.42 ± 9.16	52.69 ± 11.70	64.10 ± 3.16	72.53 ± 11.88	84.15 ± 8.76	**89.41 ± 8.39**	69.14 ± 11.70	51.30 ± 1.39	88.44 ± 7.40
13	70.69 ± 6.79	93.36 ± 2.76	72.76 ± 8.80	49.93 ± 16.01	78.19 ± 9.53	95.52 ± 0.16	85.52 ± 8.58	**96.34 ± 1.52**	79.82 ± 2.64
14	84.92 ± 6.05	**99.34 ± 0.57**	85.82 ± 9.30	92.75 ± 3.46	80.24 ± 3.46	96.83 ± 2.46	94.94 ± 5.17	90.26 ± 2.91	98.58 ± 1.26
15	88.43 ± 4.65	99.69 ± 0.39	84.82 ± 9.40	94.76 ± 4.18	97.47 ± 1.11	**99.66 ± 0.15**	99.02 ± 0.88	93.62 ± 4.57	98.68 ± 0.64
OA	74.55 ± 1.40	75.63 ± 1.86	74.88 ± 1.58	73.61 ± 1.69	75.27 ± 1.34	75.96 ± 2.43	76.78 ± 2.67	76.11 ± 0.46	**77.85 ± 1.17**
AA	76.18 ± 1.48	79.21 ± 1.58	76.23 ± 2.34	75.36 ± 2.09	77.30 ± 1.04	79.37 ± 1.90	79.59 ± 2.69	78.79 ± 0.49	**80.55 ± 1.05**
Kappa	72.51 ± 1.51	73.70 ± 2.01	72.85 ± 1.71	71.47 ± 1.85	73.26 ± 1.44	74.03 ± 2.62	74.94 ± 2.88	74.20 ± 0.50	**76.07 ± 1.27**

TABLE VI
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE SV DATASET

No.	GiaFSL	ECSA	GCC-FSL	ADAFSL	MLPA	TEFSL	CTF-SSCL	FDFSL	HOT-FSL
1	96.45 ± 1.31	96.15 ± 2.12	96.10 ± 1.13	97.60 ± 0.80	**98.96 ± 1.10**	95.89 ± 3.20	98.31 ± 1.76	96.10 ± 2.77	98.57 ± 1.95
2	98.90 ± 0.51	97.02 ± 1.59	99.25 ± 0.27	99.70 ± 0.28	97.73 ± 2.91	91.51 ± 8.51	98.21 ± 1.44	98.55 ± 1.25	**99.96 ± 0.03**
3	95.67 ± 0.34	93.31 ± 3.19	92.99 ± 1.96	98.53 ± 1.58	97.28 ± 2.09	95.69 ± 2.88	96.43 ± 2.60	77.40 ± 13.63	**99.95 ± 0.02**
4	99.67 ± 0.13	98.66 ± 1.80	99.47 ± 0.19	98.42 ± 1.46	99.48 ± 0.17	97.94 ± 1.35	**99.61 ± 0.31**	99.11 ± 0.25	96.82 ± 4.09
5	92.79 ± 2.19	89.78 ± 3.10	93.62 ± 1.81	93.84 ± 0.85	90.56 ± 3.01	85.62 ± 6.84	87.24 ± 2.69	**95.41 ± 0.77**	80.31 ± 2.71
6	99.96 ± 0.04	98.54 ± 1.03	**99.94 ± 0.03**	99.74 ± 0.32	99.48 ± 0.63	99.77 ± 0.29	99.65 ± 0.69	98.78 ± 0.27	98.62 ± 1.04
7	99.13 ± 0.54	99.05 ± 1.43	99.40 ± 0.24	97.81 ± 1.18	98.03 ± 1.71	98.90 ± 0.81	99.65 ± 0.18	**99.98 ± 0.03**	97.78 ± 1.72
8	69.98 ± 7.96	69.32 ± 6.03	59.78 ± 9.29	68.81 ± 14.56	77.28 ± 5.15	66.70 ± 7.68	78.40 ± 7.29	74.42 ± 3.77	**82.09 ± 2.56**
9	96.56 ± 2.00	98.50 ± 1.29	98.69 ± 1.62	98.65 ± 1.71	98.53 ± 1.18	98.55 ± 2.71	99.48 ± 0.48	99.41 ± 0.54	**99.69 ± 0.38**
10	83.34 ± 4.68	77.63 ± 7.32	84.73 ± 4.34	**86.68 ± 9.43**	86.36 ± 2.79	77.53 ± 6.44	84.16 ± 6.39	60.62 ± 13.36	85.37 ± 5.40
11	98.33 ± 1.28	93.94 ± 3.65	98.91 ± 0.63	96.80 ± 2.73	**99.34 ± 0.52**	97.19 ± 1.67	96.46 ± 2.73	97.69 ± 1.49	91.93 ± 6.20
12	99.56 ± 0.59	99.40 ± 0.34	99.33 ± 0.58	99.64 ± 0.34	99.47 ± 0.70	**99.69 ± 0.34**	98.27 ± 1.90	98.62 ± 1.24	92.93 ± 3.13
13	96.49 ± 1.46	**99.80 ± 0.16**	96.53 ± 0.67	98.94 ± 0.88	98.79 ± 0.94	98.44 ± 1.65	98.84 ± 1.25	95.92 ± 2.54	96.97 ± 3.82
14	96.77 ± 1.74	95.38 ± 2.94	95.83 ± 2.11	98.75 ± 0.77	99.27 ± 0.54	97.01 ± 2.99	96.57 ± 2.67	**99.47 ± 0.48**	85.22 ± 14.25
15	66.92 ± 8.51	73.39 ± 8.62	72.66 ± 2.22	76.89 ± 2.95	69.11 ± 15.17	71.36 ± 14.02	66.13 ± 16.92	**77.62 ± 4.80**	73.66 ± 5.40
16	96.57 ± 2.15	90.06 ± 7.19	95.18 ± 1.69	95.56 ± 3.84	90.46 ± 4.14	85.36 ± 6.98	94.16 ± 4.38	91.83 ± 4.11	**98.73 ± 0.72**
OA	86.82 ± 0.95	86.63 ± 0.96	85.69 ± 1.69	88.49 ± 3.01	88.79 ± 1.95	85.29 ± 1.24	88.51 ± 1.22	87.35 ± 1.83	**89.65 ± 0.58**
AA	92.94 ± 0.38	91.87 ± 0.96	92.65 ± 0.47	**94.15 ± 1.20**	93.76 ± 0.97	91.07 ± 1.04	93.22 ± 1.28	91.31 ± 1.27	92.41 ± 0.82
Kappa	85.38 ± 1.02	85.17 ± 1.06	84.17 ± 1.84	87.25 ± 3.32	87.54 ± 2.18	83.68 ± 1.40	87.21 ± 1.40	85.98 ± 2.02	**88.49 ± 0.64**

Table III reports the classification results on the PU dataset. As shown, the proposed HOT-FSL achieves the best overall performance in terms of OA, AA, and kappa. Moreover, HOT-FSL attains the highest classification accuracy on two classes (Classes 3 and 7), indicating its effectiveness in learning discriminative high-order feature representations for certain land-cover categories. This advantage enables the model to better capture class-specific structural characteristics and improve class-level recognition. Table IV reports the classification performance on the IP dataset. It can be observed that HOT-FSL consistently outperforms all competing methods across all evaluation metrics. Compared with the second-best results, HOT-FSL improves OA, AA, and kappa by 1.25%, 2.87%, and 1.92%, respectively. In addition, HOT-FSL achieves the highest classwise accuracy in eight classes, demonstrating its strong class-level discrimination capability. Table V presents the results on the HU dataset. As shown, the proposed method maintains a clear performance advantage over the competing approaches, achieving improvements of 1.07% in OA, 0.96% in AA, and 1.13% in kappa compared with the second-best method. These results suggest that HOT-FSL is capable of effectively modeling complex land-cover boundaries, leading to more discriminative feature representations. Table VI reports the classification accuracies of all methods on the SV dataset. Overall, HOT-FSL achieves competitive performance, obtaining OA and kappa values of 89.65% and 88.49%, respectively, which are superior to the suboptimal results. However, the relatively lower AA indicates that the recognition accuracy for certain minority classes remains limited, resulting in a reduced average per-class accuracy. Overall, HOT-FSL consistently attains the best or competitive performance across the four datasets.

2) Visualization Analysis: To further qualitatively evaluate the classification performance of different methods, the classification maps obtained on the four datasets are visualized and compared. As shown in Figs. 11–14, the proposed HOT-FSL produces classification maps with clearer spatial structures and fewer misclassified regions compared with the

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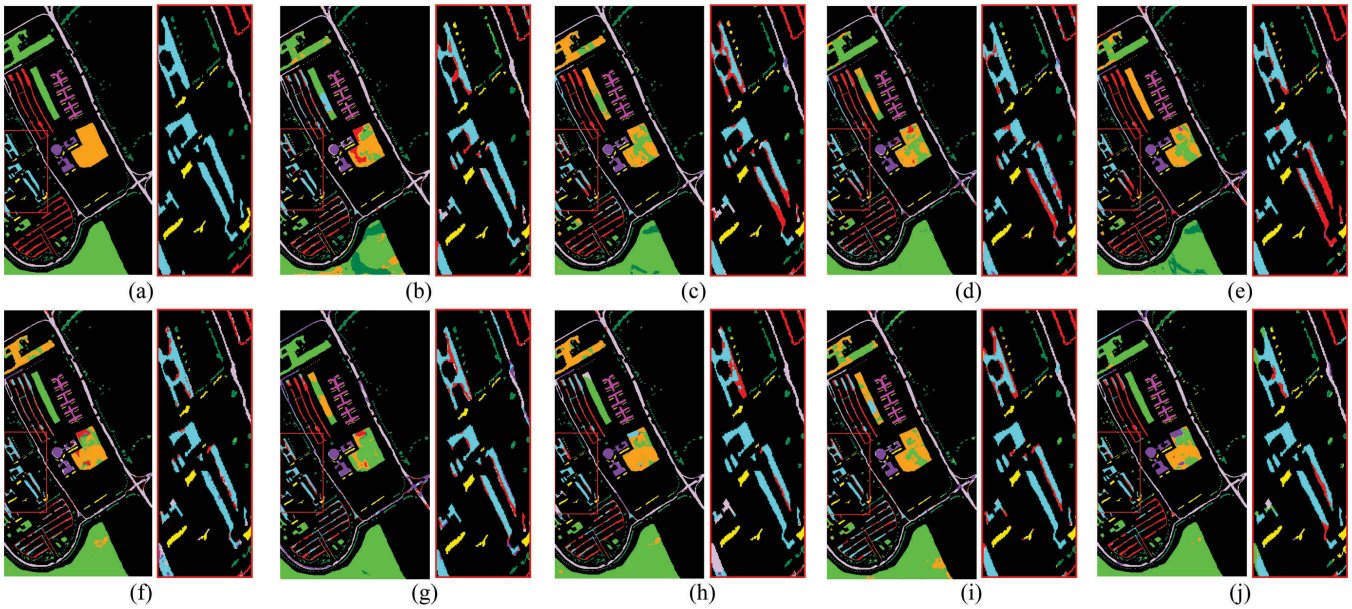


Fig. 11. Classification maps obtained via various algorithms for the PU dataset. (a) Ground truth. (b) GiaFSL. (c) ECSA. (d) GCC-FSL. (e) ADAFSL. (f) MLPA. (g) TEFSL. (h) CTF-SSCL. (i) FDFSL. (j) HOT-FSL. The regions enclosed by the red boxes are magnified right.

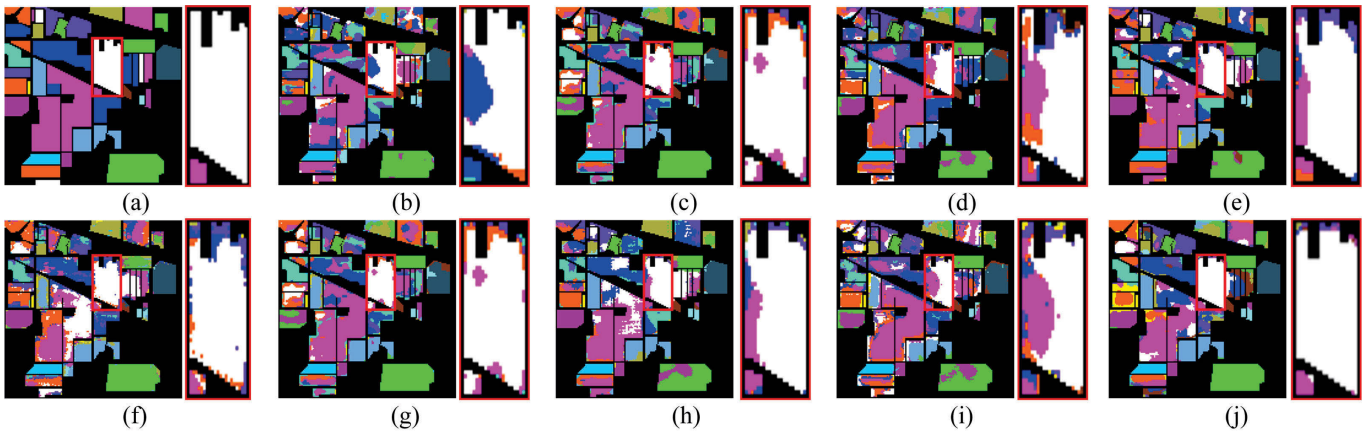


Fig. 12. Classification maps obtained via various algorithms for the IP dataset. (a) Ground truth. (b) GiaFSL. (c) ECSA. (d) GCC-FSL. (e) ADAFSL. (f) MLPA. (g) TEFSL. (h) CTF-SSCL. (i) FDFSL. (j) HOT-FSL. The regions enclosed by the red boxes are magnified right.

competing methods. On the PU and IP datasets, HOT-FSL yields more homogeneous classification regions and more accurate delineation of land-cover areas, effectively suppressing salt-and-pepper noise that commonly appears in few-shot classification scenarios. In particular, for several challenging categories with similar spectral characteristics, HOT-FSL demonstrates improved spatial consistency and reduced confusion among neighboring classes. For the HU dataset, which contains complex land-cover distributions and intricate boundary structures, the classification map generated by HOT-FSL preserves finer boundary details while maintaining region-level smoothness. This observation indicates that the proposed method can effectively capture class-specific structural information, leading to more reliable spatial predictions. On the SV dataset, although certain minority classes remain challenging, HOT-FSL still produces visually coherent classification results with fewer fragmented regions compared to other methods. Overall, the visualization results are consistent with the quantitative evaluations, further validating the effectiveness of HOT-FSL in learning discriminative representations for few-shot HSIC.

3) Sample Sensitivity Analysis: To further validate the effectiveness of the proposed HOT-FSL, a sample sensitivity analysis is conducted. Specifically, for each method, one to five samples per class are randomly selected on four different datasets, and each experiment is repeated ten times to ensure fairness and statistical reliability. The hyperparameter settings are kept consistent with those used in the previous experiments. The experimental results are illustrated in Fig. 15. Overall, the proposed HOT-FSL achieves the best performance under the five-shot setting. On the PU dataset, the classification accuracy of all methods exhibits a positive correlation with the number of training samples, and HOT-FSL attains the highest accuracy when five samples per class are used. Notably, on

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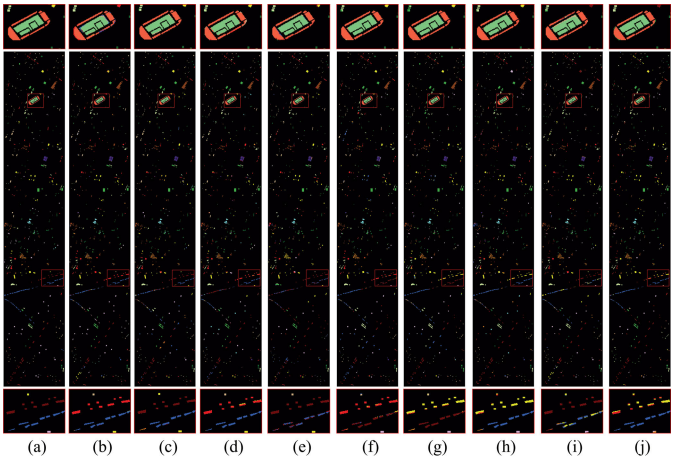


Fig. 13. Classification maps obtained via various algorithms for the HU dataset. (a) Ground truth. (b) GiaFSL. (c) ECSA. (d) GCC-FSL. (e) ADAFSL. (f) MLPA. (g) TEFSL. (h) CTF-SSCL. (i) FDFSL. (j) HOT-FSL. The regions enclosed by the red boxes are magnified above and below.

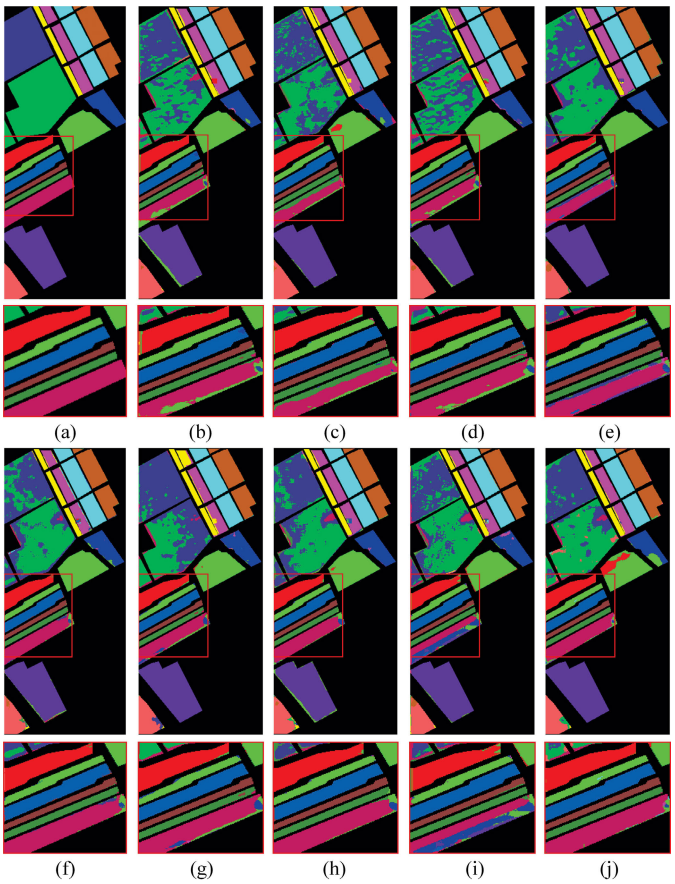


Fig. 14. Classification maps obtained via various algorithms for the SV dataset. (a) Ground truth. (b) GiaFSL. (c) ECSA. (d) GCC-FSL. (e) ADAFSL. (f) MLPA. (g) TEFSL. (h) CTF-SSCL. (i) FDFSL. (j) HOT-FSL. The regions enclosed by the red boxes are magnified below.

the IP, HU, and SV datasets, HOT-FSL performs slightly worse than some competing methods when the number of samples per class is less than or equal to two. However, when the sample number exceeds three, the performance of HOT-FSL increases rapidly with the growing number of samples, eventually achieving the best results under the five-shot setting. This is mainly due to the difficulty of reliably estimating high-order spatial–spectral representations from very limited samples, which makes class prototypes more sensitive to noise. In addition, the prototype refinement in CTPUT may further propagate this uncertainty when the initial samples are insufficient. Notably, statistical analysis shows that the performance differences under extremely low-shot settings (e.g., one-shot or two-shot) are not significant ($p > 0.05$), suggesting that the observed fluctuations are mainly caused by estimation instability rather than consistent degradation. As the number of samples increases, the performance becomes more stable and consistent, demonstrating the effectiveness of the proposed high-order modeling. Overall, following the commonly adopted setting where five samples are randomly selected for each class, the proposed method demonstrates highly competitive performance.

TABLE VII
ABLATION STUDY OF DIFFERENT COMPONENTS

	HOTE	PCGI	CTPUT	PU	IP	HU	SV
case 1	×	×	×	81.53	68.63	74.73	86.93
case 2	√	×	×	83.58	70.22	76.87	88.90
case 3	√	×	√	84.51	70.46	76.16	89.16
case 4	×	√	×	82.18	68.69	74.36	84.24
case 5	√	√	×	84.32	70.84	77.40	89.36
case 6	√	√	√	**84.97**	**71.10**	**77.85**	**89.65**
case 7	Axial	√	√	83.84	70.25	76.12	89.16
case 8	√	Pro	√	83.02	69.06	74.77	86.45

TABLE VIII
ABLATION STUDY ON DIFFERENT DECOMPOSITION STRATEGIES WITHIN THE HOTE MODULE

Method	PU	IP	HU	SV
Additive	84.37±1.32	70.20±1.06	77.10±1.24	88.87±1.25
Multiplicative	**84.97±1.73**	**71.10±2.17**	**77.85±1.17**	**89.65±0.58**

E. Ablation Experiments

To evaluate the contribution of each module, we perform ablation studies in this subsection, focusing primarily on the HOT-FSL component. Specifically, we consider the following four experimental settings.

Case 1: The proposed network excludes HOTE, PCGI, and CTPUT. Following prior works [45], [48], a 3-D-ResNet is adopted as the feature extractor.

Case 2: The proposed network includes only the HOTE module.

Case 3: The proposed network incorporates HOTE and PCGI while excluding CTPUT.

Case 4: The proposed network incorporates HOTE and CTPUT while excluding the PCGI module.

Case 5: The proposed network retains only the PCGI module while removing both HOTE and CTPUT.

Case 6: The full HOT-FSL architecture.

Case 7: The proposed network replaces the HOTE with axial decomposed attention [54].

Case 8: The proposed network replaces the PCGI with simple prototype metric learning.

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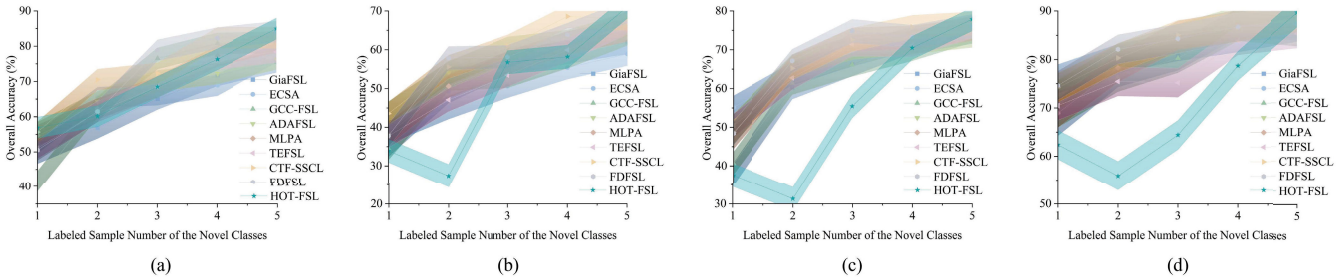


Fig. 15. OA (%) of various methods on the four data sets with different numbers of labeled samples. (a) PU. (b) IP. (c) HU. (d) SV.

TABLE IX
TRAINING TIME, TESTING TIME, AND PARAMETER COMPARISON OF DIFFERENT METHODS

Datasets	Metric	Methods								
		GiaFSL	ECSA	GCC-FSL	ADAFSL	MLPA	TEFSL	CTF-SSCL	FDFSL	HOT-FSL
PU	Train time(s)	899.60	558.29	642.80	6823.63	423.81	289.00	138.12	446.59	525.05
	Test time(s)	4.63	4.88	4.61	4.94	7.75	18.35	4.53	12.21	12.44
	Params(M)	0.19	0.17	0.19	0.04	0.05	0.92	0.08	0.18	0.51
IP	Train time(s)	1281.67	973.15	1244.58	7341.35	356.74	346.87	203.94	345.49	566.42
	Test time(s)	1.55	1.14	1.29	1.41	1.33	2.95	1.19	2.96	9.37
	Params(M)	0.20	0.17	0.21	0.05	0.06	0.96	0.09	0.19	0.52
HU	Train time(s)	1687.71	947.18	901.08	3942.10	369.63	273.17	205.94	539.20	449.01
	Test time(s)	2.54	1.20	1.85	1.30	1.86	3.98	1.32	3.00	2.86
	Params(M)	0.19	0.17	0.19	0.04	0.05	0.94	0.08	0.18	0.51
SV	Train time(s)	1344.50	1307.14	1299.30	6206.61	263.20	385.26	265.07	703.21	585.14
	Test time(s)	7.05	5.52	10.24	8.29	3.91	21.64	5.31	23.63	11.87
	Params(M)	0.21	0.19	0.21	0.05	0.06	0.98	0.09	0.19	0.52

The ablation study results are reported in Table VII, which quantitatively evaluates the contribution of each component in the proposed HOT-FSL framework. As shown in Case 1, when all the proposed modules are removed and a 3-D-ResNet is employed as the backbone feature extractor, the classification performance is relatively limited across all target datasets. This observation indicates that conventional feature extractors struggle to effectively capture discriminative representations under cross-domain few-shot settings. By introducing the HOTE module alone (Case 2), the performance is consistently improved on all datasets, demonstrating the effectiveness of high-order spatial–spectral modeling in enhancing feature discrimination. In Case 3, further incorporating the PCGI module yields additional performance gains, which verifies the benefit of injecting prototype-conditioned priors for alleviating cross-domain distribution discrepancies. Case 4 achieves the best performance on all datasets by integrating HOTE, PCGI, and CTPUT, confirming that these components are complementary and jointly contribute to the overall effectiveness of HOT-FSL. It can be observed from Case 5 that replacing the HOTE module with axial attention leads to consistent performance degradation across all datasets. This indicates that, compared with the proposed method, the sequential factorization strategy is insufficient to effectively capture the intrinsic spatial–spectral dependencies. Case 6 shows that replacing PCGI with simple prototype-based metric learning consistently degrades performance, highlighting the necessity of explicit support-query interaction. Overall, the complete model consistently achieves superior performance across all four datasets.

To justify the necessity of the proposed decomposition strategy and clarify its difference from conventional factorized attention, we conduct an ablation study within the HOTE module. Table VIII presents the results of the additive formulation (sum) and the proposed multiplicative formulation (product). Compared with the additive strategy, the proposed multiplicative strategy consistently yields performance improvements of 0.60%, 0.90%, 0.75%, and 0.78% on the PU, IP, HU, and SV datasets, respectively. Although the performance gain is moderate, the improvement is consistent across all datasets, demonstrating the robustness of the proposed decomposition strategy. This difference mainly stems from the distinct modeling capabilities of the two formulations. Specifically, the additive strategy aggregates independently computed directional attention responses, which lack explicit cross-dimensional interaction and, thus, limit their representational capacity. In contrast, the multiplicative formulation introduces a sequential dependence between spatial dimensions, where the attention output of one direction is propagated to the next, effectively capturing cross-dimensional correlations. Therefore, the proposed formulation goes beyond conventional factorized attention and provides a more expressive approximation to high-order

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spatial dependencies, validating the necessity of the adopted decomposition strategy.

F. Complexity Analysis

We further evaluate the computational efficiency of HOT-FSL in terms of training time, inference time, and parameter size, as summarized in Table IX. Despite introducing high-order representation modeling and transductive prototype refinement, HOT-FSL maintains competitive computational cost compared with existing few-shot HSIC methods. From the perspective of training time, the proposed method operates within a reasonable range and is substantially more efficient than ADAFSL, which incurs a notably higher training cost. The ADAFSL method increases computational cost by employing a dual-classifier strategy that doubles per-sample classification, along with additional element-wise weighting and dynamic gradient modulation in each iteration. In terms of inference time, HOT-FSL clearly outperforms TEFSL, another transductive few-shot method, indicating that the proposed CT optimization introduces favorable efficiency during prediction. Regarding parameter size, although HOT-FSL is not the smallest model among the compared methods, its scale remains within an acceptable range for practical deployment. Overall, HOT-FSL achieves a well-balanced tradeoff between classification accuracy and computational efficiency, demonstrating strong comprehensive performance.

IV. Conclusion

This article presents a novel HOT-FSL framework for cross-domain few-shot HSIC, comprising three key components: a HOTE, a PCGI, and a CTPUT. Specifically, motivated by Kronecker decomposition, HOTE is developed to capture the intrinsic high-order spatial–spectral representations embedded in hyperspectral imagery. PCGI introduces class-level priors from the support set into query features in a query-adaptive manner, thereby enhancing feature discriminability and alleviating prototype bias under limited supervision. Moreover, CTPUT performs reliable transductive prototype calibration by softly transferring query statistics through CT, avoiding the need for hard pseudolabel assignments. Although the proposed method demonstrates consistent performance gains across multiple benchmark datasets, there remains room for improvement under extremely low-shot scenarios (e.g., one-shot and two-shot), where limited supervision poses significant challenges. In addition, the underlying theoretical properties of high-order representations warrant further investigation, and more intuitive visualizations could be explored to enhance interpretability. Meanwhile, the current framework introduces additional computational overhead due to its sophisticated design. Future work will, therefore, focus on strengthening the theoretical foundation, improving data efficiency in ultralow-shot settings, and developing more efficient and scalable architectures for practical application, such as lightweight attention mechanisms and knowledge distillation strategies to reduce inference overhead.

References

[1] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[2] S. Lu et al., "Vision foundation models in remote sensing: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 3, pp. 190–215, Sep. 2025.

[3] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[4] M. J. Khan, H. S. Khan, A. Yousaf, K. Khurshid, and A. Abbas, "Modern trends in hyperspectral image analysis: A review," *IEEE Access*, vol. 6, pp. 14118–14129, 2018.

[5] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[6] V. Kumar, R. S. Singh, M. Rambabu, and Y. Dua, "Deep learning for hyperspectral image classification: A survey," *Comput. Sci. Rev.*, vol. 53, Aug. 2024, Art. no. 100658.

[7] B. G. Ram, P. Oduor, C. Igathinathane, K. Howatt, and X. Sun, "A systematic review of hyperspectral imaging in precision agriculture: Analysis of its current state and future prospects," *Comput. Electron. Agricult.*, vol. 222, Jul. 2024, Art. no. 109037.

[8] C. Weber et al., "Hyperspectral imagery for environmental urban planning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1628–1631.

[9] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2014.

[10] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.

[11] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017.

[12] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[13] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[14] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[15] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[16] W. Li, C. Chen, M. Zhang, H. Li, and Q. Du, "Data augmentation for hyperspectral image classification with deep CNN," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 593–597, Apr. 2019.

[17] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[18] A. Dosovitskiy, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[19] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[20] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[21] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415.

[22] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. 1st Conf. Lang. Modeling*, 2024.

[23] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-mamba: 3D-spectral–spatial mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.

[24] Y. Chu, J. Cao, J. Xia, and W. Ding, "TGMN: Two-stage graph convolutional mamba network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 37, no. 6, pp. 2914–2928, Jun. 2026, Art. no. 3642545.

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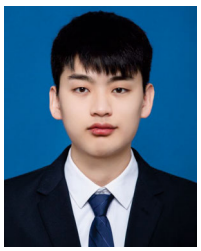
[25] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.
[26] C. Zheng, N. Wang, and J. Cui, "Hyperspectral image classification with small training sample size using superpixel-guided training sample enlargement," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7307–7316, Oct. 2019.
[27] W. Hou, N. Chen, J. Peng, and W. Sun, "A prototype and active learning network for small-sample hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, Art. no. 5510805.
[28] X. Yang, Z. Song, I. King, and Z. Xu, "A survey on deep semi-supervised learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 9, pp. 8934–8954, Sep. 2023.
[29] A. Jaiswal, A. R. Babu, M. Z. Zadeh, D. Banerjee, and F. Makedon, "A survey on contrastive self-supervised learning," *Technologies*, vol. 9, no. 1, p. 2, Dec. 2020.
[30] A. Sellami, M. Farah, I. Riadh Farah, and B. Solaiman, "Hyperspectral imagery classification based on semi-supervised 3-D deep neural network and adaptive band selection," *Expert Syst. Appl.*, vol. 129, pp. 246–259, Sep. 2019.
[31] H. Wu and S. Prasad, "Semi-supervised deep learning using pseudo labels for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1259–1270, Mar. 2018.
[32] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501813.
[33] J. Feng, N. Zhao, R. Shang, X. Zhang, and L. Jiao, "Self-supervised divide-and-conquer generative adversarial network for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536517.
[34] T. Hospedales, A. Antoniou, P. Micaelli, and A. Storkey, "Meta-learning in neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5149–5169, Sep. 2022.
[35] Y. Song, T. Wang, P. Cai, S. K. Mondal, and J. P. Sahoo, "A comprehensive survey of few-shot learning: Evolution, applications, challenges, and opportunities," *ACM Comput. Surv.*, vol. 55, no. 13s, pp. 1–40, Dec. 2023, Art. no. 271.
[36] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022, Art. no. 3192712.
[37] D. Song, F. Qin, B. Wang, and S. Dai, "Few-shot learning with label smoothing and metric space optimization for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535722.
[38] Y. Wang et al., "Heterogeneous few-shot learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, Art. no. 5510405.
[39] H. Xu, S. Zhi, S. Sun, V. Patel, and L. Liu, "Deep learning for cross-domain few-shot visual recognition: A survey," *ACM Comput. Surv.*, vol. 57, no. 8, pp. 1–37, Apr. 2025, Art. no. 215.
[40] Z. Ye, J. Wang, H. Liu, Y. Zhang, and W. Li, "Adaptive domain-adversarial few-shot learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532017.
[41] Y. Cheng, W. Zhang, H. Wang, and X. Wang, "Causal meta-transfer learning for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521014.
[42] S. Feng, H. Zhang, B. Xi, C. Zhao, Y. Li, and J. Chanussot, "Cross-domain few-shot learning based on decoupled knowledge distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534414.
[43] B. Xi et al., "Semisupervised cross-scale graph prototypical network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9337–9351, Nov. 2023.
[44] Z. Ye, J. Wang, T. Sun, J. Zhang, and W. Li, "Cross-domain few-shot learning based on graph convolution contrast for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504614.
[45] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1912–1925, Feb. 2024.
[46] Y. Liu et al., "Learning to propagate labels: Transductive propagation network for few-shot learning," 2018, *arXiv:1805.10002*.
[47] L. Tian et al., "Prototypes-oriented transductive few-shot learning with conditional transport," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16271–16280.
[48] B. Xi et al., "Transductive few-shot learning with enhanced spectral–spatial embedding for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 34, pp. 854–868, 2025.
[49] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Space Appl. Lab., Univ. Tokyo, Tokyo, Japan, Tech. Rep. SAL-2016-05-27, 2016, vol. 5, p. 5.
[50] Y. Wang, F. Zhang, J. Tian, X. Gong, and Z. Li, "An entropy-driven clustering and semantic association framework for cross-domain few-shot hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025, Art. no. 5506105.
[51] H. Liu, J. He, Y. Li, and Y. Bi, "Multilevel prototype alignment for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4400115.
[52] B. Xi, Y. Zhang, J. Li, Y. Li, Z. Li, and J. Chanussot, "CTF-SSCL: CNN-transformer for few-shot hyperspectral image classification assisted by semisupervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532617.
[53] B. Qin, S. Feng, C. Zhao, W. Li, R. Tao, and W. Xiang, "Cross-domain few-shot learning based on feature disentanglement for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514215.
[54] F. Zhu, C. Shi, K. Shi, and L. Wang, "Joint classification of hyperspectral and LiDAR data using hierarchical multimodal feature aggregation-based multihead axial attention transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 3533475.



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